

reviewer

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Multimedia

- Multimedia is something people can see, hear, touch and read on websites, radio, television, and etc.
- It can be defined as a combination of picture, text, sound, video and animation.
- The combination of more than one media.

- **Interactive Multimedia** - When a user is allowed to control the elements of multimedia
- **Hypermedia** - When the user is given the structure of linked elements to control it.

It is a combination of text, graphic, sound, animation, and video that is delivered interactively to the user by electronic or digitally manipulated means.

Multi

- Many, Multiple

Media

- Tools that is used to represent or do a certain things, delivery medium, a form of mass communication - newspaper, magazine / tv.
- Distribution tool & information presentation - text, graphic, voice, images, music and etc.

- It is a mixture of different forms of media
- Provides important information in an interactive way.

COMBINING CONTENT FROM VARIOUS MEDIA

- Text and typography
- Coloured, pictorial or graphical backgrounds
- Images and graphics such as photographs, illustrations, cartoons, diagrams, shapes
- arrows, lines, etc
- 2D and 3D animation
- Audio files such as a song, sound effects, music, etc.
- Video files such as a short movie clip, advertisement

multimedia

FEATURES OF MULTIMEDIA

1 TEXT MEDIA

- A broad term for something that contains words to express something.
- Most basic element of multimedia
- Could help convey the intended message to the users

2 GRAPHICS MEDIA

- Two-dimensional figure or illustration
- Could be produced manually or by computer graphics technology
- To show more clearly what a particular information is all about

3 AUDIO MEDIA

- Produced by vibration, as perceived by the sense of hearing.
- Could come in the form of speech, sound effects and also music score.

4 ANIMATION MEDIA

- The illusion of motion created by the consecutive display of images of static elements.
- Used to further enhance / enriched the experience of the user to further understand the information conveyed to them.

5 VIDEO MEDIA

- Is the technology of capturing, recording, processing, transmitting, and reconstructing moving pictures.
- More towards photo realistic image sequence / live recording
- Takes a lot of storage space.

6 VIRTUAL REALITY

HYPERMEDIA

A combination of hypertext, graphics, audio, video, (linked elements) and interactivity culminating in a complete, non-linear computer-based experience.

CATEGORIES OF MULTIMEDIA

LINEAR MULTIMEDIA

- Information is read or viewed in a continuous sequence
- Begin at a predetermined starting point and end at a predetermined end point
- Can be automated so that each screen comes after a fixed time interval
- User have no control over the content that is being showed to them; not interactive
- Ex. ppt presentation, movie, lecture

NON-LINEAR MULTIMEDIA

- Information is not presented in sequential or chronological manner
- Interactive and require audience interaction
- Users have control over the content that is being showed to them
- Users are given navigational control
- Ex. Web, games, courseware, interactive cd

multimedia

APPLICATIONS OF MULTIMEDIA

BUSINESS

- Multimedia is used for advertising and selling products on the Internet.
- Ex. for training or educating staff members
- Multimedia is one of the best ways to provide short-term training to the workers in a company.

RESEARCH AND MEDICINE

- It is increasingly used in research in the fields of science, medicine and mathematics.
- For modelling and simulation

PUBLIC ACCESS NOTES

- An area of application where several multimedia applications will be available very soon
- Ex. Tourist Information System

ENTERTAINMENT

- Used to create special effects in films, TV serials, radio shows, games and animations.
- Multimedia Games; fly aeroplanes, drive cars, etc.

INDUSTRY

- Multimedia is used to present information to all people related to the industry
- Helpful in advertising and selling products all over the world over internet

COMMERCIAL

- Creative presentations are used to grab the attention of the masses in advertising.
- Work on advanced multimedia presentations to sell ideas or make training programs more interesting

EDUCATION

- Used as a source of information in the field of education
- Pupils can research on various topics
- To make teaching more interesting and fun
- Various computer-based training (CBT) courses

MULTIMEDIA IN PUBLIC PLACES

- Multimedia will become available at stand-alone terminals to provide information and help
- Information booths on railways stations, hotels, etc.

ENGINEERING

- Use multimedia in computer simulations for anything from entertainment to training
- Multimedia for software interfaces are often done as collaboration between creative professionals and software engineers

CAUTION

Make your multimedia to the point and precise or it can have adverse effects on the business in financial terms.

multimedia

STAGES OF MULTIMEDIA

1. **Pre-production:** The process before producing multimedia project.
2. **Production:** The process in which multimedia project is produced.
3. **Post-production:** The process after the production of multimedia project.

PRE-PRODUCTION

Idea or Motivation:

- The first question the production team ask is “why” you want to develop a multimedia project

Product Concept and Project Goals:

- The production team decides what the product needs to accomplish in the market; what information and function they need to provide to meet desired goals

Target Audience:

- The production team thinks about target age groups, and how it affects the nature of the product. Background of target customers and the types of references that will be fully understood.

Delivery Medium and Authoring Tools:

- The production team decides the medium through which the information reach the audience. The production team also ascertain what authoring tools should be used in the project. A few of the authoring tools are graphics, audio, video, text, animation, etc.

Planning:

- Planning is the key to the success of most business endeavours, and this is definitely true in multimedia.
- It includes creating and finalizing flowchart and resource organization in which the product's content is arranged into groups. Timeline, roles, phases, etc.

PRODUCTION

- All components of planning come into effect.
- During this phase graphic artists, instructional designers, animators, audiographers and videographers begin to create artwork, animation, scripts, video, audio and interface.
- Runs easily if project manager delegates tasks properly

Scriptwriting

- The scripts for the text, transitions, audio narrations, voice-overs and video are written

Art

- Illustrations, graphics, buttons, and icons are created using the prototype screens as a guide. Existing photographs, illustrations, and graphics are digitized for use in an electronic medium.

3D Modelling and Animation

- The 3D artwork is created, rendered, and then prepared for use in the authoring tool. The 3D animations require their own storyboards and schedules.

Authoring

- All the pieces come together in the authoring tool.
- Functionality is programmed, and 2D animation is developed.
- The final working product is created

Shooting and Recording Digitizing Video

- The edited scripts are used to plan the budget, performers, time schedules and budget, then the shoot is scheduled followed by recording

Quality Control

- Goes on throughout the process
- The main goal of production is to make the next stage, post production, run smoothly and flawlessly.

POST-RODUCTION

- After the production of the multimedia project, post-production technicalities should be addressed to produce a perfect and error free project.
- It is one of the most fundamental of all stages of production.

Testing

- The product is tested on multiple computers and monitors. It is imperative to evaluate, test and revise the product to make sure the quality and success of the product.

Mastering

- Mastering can be as simple as writing a CD-ROM or floppy disk. Or it can be as complex as sending the files to a service that will create a pre-master from which the master is made.

Archiving and Duplication Notes

- The original files, including audio, video, and the native software formats, are archived for future upgrades or revisions. The duplicates are created from the original and packaged accordingly.

Marketing and Distribution

- Marketing is significant to the success of a product. The survival of a company and its products depends greatly on the product reaching the maximum number of audience.

text media

TEXT

- Text is an important component used in many multimedia applications.
- Used to create words, sentences and paragraphs
- Source of information
- Good at providing basic information
- Text comes in any forms of words, and symbols
- Most common system of communication: **SPOKEN** and **WRITTEN**
- A piece of writing that you read or create

FONTS AND FACES

TYPEFACE - a family of graphic characters that usually includes many types of sizes and styles

FONT - Is a collection of characters of a single size and style that belongs to a particular typeface family.

EXAMPLES OF CASES

- **UPPERCASE**
- **LOWERCASE**
- **PROPERCASE**

EXAMPLES OF FONTS

- **BOLD**
- **ITALIC**
- **UNDERLINE**
- **OUTLINE**

TYPES OF TEXT

EXPOSITORY TEXTS

- It is intended to identify and characteristics, experience, facts, situations and actions in either abstract or real element
- Meant to explain, inform or describe and they are the most frequently use to write structures.

NARRATIVE TEXTS

- Entertains, instructs or inform readers by telling a story
- Deal with imaginary or real world
- Can be **FICTIONAL** and **NON-FICTIONAL**

FICTIONAL

- Fairy Tales
- Myths
- Novels
- Science Fiction
- Horror or Adventure Stories
- Fable
- Legends

NON-FICTIONAL

- Articles
- Newspapers
- Historical Writings

ARGUMENTATIVE TEXTS

- Aims to change the readers beliefs
- Often contain negative qualities or characteristics of something/someone
- Try to persuade their readers that an object, product, idea is in some way better than the other

HISTORY OF MULTIMEDIA TEXT

MULTIMEDIA TEXT

- Refers to text integrated with other media

1

EARLY TEXT AND ILLUSTRATIONS (BEFORE 20TH CENTURY)

ANCIENT MANUSCRIPTS AND ILLUMINATED TEXTS (BEFORE 1500s)

Medieval manuscripts included decorated letters and images alongside text to enhance understanding.

EARLY PRINT MEDIA (1500s-1800s)

Books and newspapers incorporated illustrations, engravings, and later, photographs.

2

PHOTOGRAPHY AND FILM INTEGRATION (19TH - EARLY 20TH CE)

PHOTOGRAPHY IN PRINT (1839 ONWARD)

Newspapers and books started combining text with photographs for a richer reading experience. S

SILENT FILMS (1890s - 1920s)

Intertitles (text between scenes) were used to convey dialogue and narration.

3

RADIO, TELEVISION, AND EARLY DIGITAL MULTIMEDIA (20TH CENTURY)

RADIO BROADCASTS (1920s - 1950s)

Although audio-based, radio programs sometimes used textual supplements in magazines.

TELEVISION (1950s - 1970s)

Text appeared in captions, news tickers, and advertisements.

HYPERTEXT (1960s- 1980s)

Early concepts of linking text digitally emerged (e.g., Ted Nelson's "hypertext" theory)

4

THE DIGITAL REVOLUTION (1990s - 2000s)

HYPERTEXT AND THE WEB (1990s)

The invention of HTML and the World Wide Web enabled interactive text linking with images, video, and sound.

MULTIMEDIA CD-ROMs (1990s)

Educational and entertainment content used interactive text, images, and audio.

FLASH AND INTERACTIVE WEBSITES (2000s)

Adobe Flash allowed animated and interactive text elements on web pages.

5

MODERN MULTIMEDIA TEXT (2010s - PRESENT)

SOCIAL MEDIA AND INFOGRAPHICS

Platforms like Instagram and TikTok blend text with images and videos for storytelling.

AUGMENTED AND VIRTUAL REALITY (AR/VR)

Text appears as overlays in immersive experiences.

audio media

SOUND DESIGN

- The process of creating, manipulating, and integrating sound into multimedia projects.
- Involves crafting soundtracks, sound effects, and audio environments.

IMPORTANCE

- Enhance storytelling, sets the mood, and directs audience emotions
- Ensures coherence between sound and visual elements

HISTORY OF AUDIO PLAYERS

- **PHONOGRAPH** (Developed in 1855)
- **PHONOGRAPH** (Patented in 1887)
- **RADIO** (1920s)
- **8-TRACK**
- **CASSETTE TAPES**
- **COMPACT DISC**
- **DIGITAL AUDIO TAPE** (1987)
- **MINI DISC** (1993)
- **MP3** (Invented in 1989)
- **AAC - MP4** (2001)

THE NATURE OF SOUND

FREQUENCY

- The difference in time between the formations of two crests is termed as the period
- Measured in seconds
- **PEAKS or CRESTS**
- The number of peaks that occur in one second is the frequency
- **PITCH** - another term associated with frequency
- If an object oscillates rapidly, it creates a "**high-pitched**" sound. A **low-frequency sound** on the other hand is produced by an object that vibrates slowly
- Frequency is measured by the **number of cycles (vibrations) per second** and the unit of frequency is **hertz (Hz)**.
- The number of waves passing a point in one second
- Human Ear (20-20000 Hz)

THE NATURE OF SOUND

WAVELENGTH

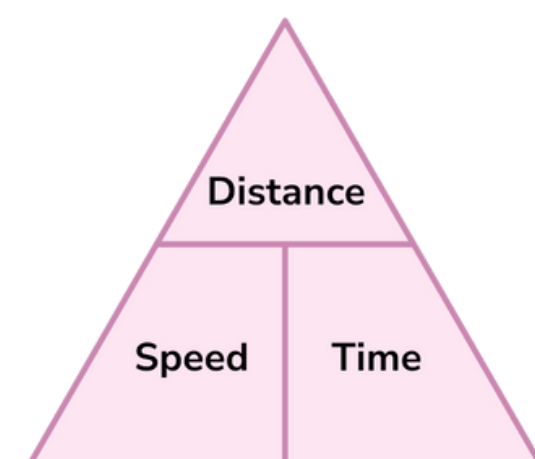
- The distance from the midpoint of one crest to the midpoint of the next crest.
- It is represented by the symbol λ

DOPPLER EFFECT

- **Sound waves** are compressions and rarefactions of air.
- When the object making the sound is moving toward you, the frequency goes up due to the waves getting pushed more tightly together. The opposite happens when the object moves away from you and the pitch goes down.

SPEED OF SOUND

- The speed of sound is the rate at which sound waves travel through a medium
- It depends on factors such as temperature, pressure, and the type of medium (air, water, or solid).



- d = distance (in meters)
- v = speed of sound (in meters per second)
- t = time taken for the sound to travel (in seconds)

SPEED OF SOUND IN AIR

$$V = 331.3 + (0.6 \times T)$$

Medium	Speed of Sound (m/s)
Air (20°C)	343
Water	1,480
Steel	5,960

audio media

THE NATURE OF SOUND

BANDWIDTH

- The difference between the highest and the lowest frequency contained in a signal
- A signal which spans the range of 200-3200 Hz has a bandwidth (BW) of:

$$BW = 3200 - 200 = 3000\text{Hz}$$

HARMONICS

- **TIMBRE** - the combinations of frequencies generated by an instrument
- **OVERTONES or HARMONICS** - a timbre consists of a fundamental or main frequency and other minor frequencies
- **FUNDAMENTAL FREQUENCY** - A string may vibrate with the lowest frequency f
- **HARMONIC FREQUENCIES** - Possible modes of vibration whose frequencies are all multiples of the lowest frequency

VELOCITY OF SOUND

- May be found directly by measuring the time required for the waves to travel a measured distance.
- The velocity of sound increases as the temperature rises and decreases as the temperature falls.

The Speed of Sound in Various Media

Medium	Meters/second
Air	331.5
Hydrogen	1270
Carbon dioxide	258
Water	1450
Iron	5100
Glass	5500

INTRO TO DIGITAL AUDIO

SAMPLING RATE

- When sampling a sound, the computer processes snapshots of the waveform. The frequency of these snapshots is called the sampling rate.

INTRO TO DIGITAL AUDIO

DIGITIZATION

- The process of assigning a discrete value to each of the sampled values
- Performed by an Integrated Chip (IC) called an **A to D Converter**

FIDELITY

- Defined as the closeness of the recorded version to the original sound

SOUND FORMATS AND SETTINGS

- Another aspect to consider is the **bit-resolution**. Along with sampling rate, the resolution determines the accuracy with which a sound is digitized. The increase in the number of bits in a recording makes the sound playback increasingly realistic.

COMPRESSION

- Reduces data size, making file transfers faster and more efficient while lowering bandwidth usage. This ensures better sound quality, even on slow networks.
- **LOSSLESS COMPRESSION** - data are not altered or lost in the process of compression or decompression.
- **LOSSY COMPRESSION** - The loss of this data is such that the object looks more or less like the original; loss of some info; most commonly used

TYPES OF SOUND IN MULTIMEDIA

DIEGETIC SOUND

- Comes from within the story's world (characters hear it)
- Examples: Dialogue, footsteps, radio playing in a scene

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TYPES OF SOUND IN MULTIMEDIA

NON-DIEGETIC SOUND

- Added for dramatic or emotional impact (characters don't hear it).
- Examples: Background music, narration, dramatic sound effects.

AMBIENT SOUND

- Background noise that enhances realism.
- Examples:
 - Birds chirping in a nature documentary.
 - City traffic in an urban scene.
- Purpose: Establishes location, mood, and realism.

EFFECTS (SFX) SOUND

- Artificially created or enhanced sounds.
- Examples:
 - Sci-fi laser gun sounds.
 - Swooshing sounds for transitions in presentations.
- Purpose: Adds emphasis and realism.

MUSIC AND SCORES

- Musical compositions used in multimedia.
- Examples:
 - Fast, upbeat music for action scenes.
 - Soft piano for emotional moments.
- Purpose: Influences audience emotions and storytelling.

CREATING IMMERSIVE SOUNDSCAPES

LAYERING SOUNDS FOR DEPTH

- Combines multiple sound elements for a rich experience.
- Example: A war scene might include gunfire, explosions, distant shouting, and a dramatic score.

CREATING IMMERSIVE SOUNDSCAPES

SPATIAL AUDIO TECHNIQUES

- **Spatial Audio** - 3D sound positioning to enhance immersion.
- **Techniques:**
 - Binaural Recording:** Uses two microphones for a 3D effect in headphones.
 - Surround Sound:** Uses multiple speakers for directional sound.
 - Object-Based Audio:** Customizable sound positioning in digital environments.

BALANCING SOUND ELEMENTS

- Ensuring dialogue, music, and effects work together.
- Tools used:
 - Equalization (EQ)** - Adjusts sound frequencies.
 - Compression** - Controls volume consistency.
 - Panning** - Positions sound in left/right speakers.

USING MUSIC TO ENHANCE EMOTION

CHOOSING THE RIGHT GENRE

- Music influences emotional perception.
- Examples:
 - Classical music** - sophistication
 - Rock music** - excitement
 - Electronic music** - futuristic tone

TEMPO AND MOOD CORRELATION

- **Fast Tempo** - Increases energy and excitement.
- **Slow Tempo** - Evokes calmness, sadness, or suspense.

CULTURAL CONTEXT OF MUSIC

- Different cultures interpret sounds and music uniquely.

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USING MUSIC TO ENHANCE EMOTION

MUSIC LICENSING AND RIGHTS

- Legal considerations when using copyrighted music.
- Examples:
Royalty-free music libraries.
Hiring composers for original soundtracks.

VOICEOVER TECHNIQUES

SELECTING VOICE TALENT

- Choose a voice that matches the multimedia's tone.
- Examples:
Deep voice = authority
Soft voice = warmth and calmness

SCRIPTWRITING FOR VOICEOVERS

- Keep it concise, clear, and engaging.
- Include natural pauses and emphasis for better delivery.

RECORDING & EDITING BEST PRACTICES

- **Equipment:** Use quality microphones and a noise-free environment.
- **Software:** Adobe Audition, Audacity.
- **Post-Production:** Removing background noise, adjusting levels.

SOUND EDITING & MIXING FUNDAMENTALS

INTRO TO AUDIO EDITING SOFTWARE

- Popular Tools:
Adobe Audition – Professional-grade mixing.
Audacity – Free, beginner-friendly.
Pro Tools – Industry-standard for music production.

BASIC MIXING TECHNIQUES

- Adjusting levels, applying effects, balancing elements.
- Key Techniques:
Equalization (EQ)
Compression
Reverb & Delay for depth

BALANCING SOUND LEVELS

- Ensures all sound elements are clearly audible.
- Example: Making sure dialogue isn't drowned by background music.

EXPORTING AND FINALIZING AUDIO

- Choose appropriate formats (MP3, WAV).
- Check for distortions or inconsistencies before finalizing.

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GRAPHICS MEDIA

GRAPHICS

- Any visual representation of data and includes a variety of forms including drawings, photographs, line art, and etc.
- May be images only or may also include text
- Can be used for decorative or functional purposes, including educating, informing, illustrating, or entertaining
- **GRAPHIC** - relates to visual art or representation
- **MEDIA** - indicates the means of communication

ETYMOLOGY

Graphic is derived from the **Greek** word '**graphikos**,' which pertains to writing or drawing

Media comes from the **Latin** '**medium**,' meaning a means or agent through which something is conveyed.

USES OF GRAPHICS

1

- To inform, illustrate, or entertain
- To make concepts easier to understand

2

- To make content more visually appealing
- To communicate clearly and effectively

USES OF GRAPHICS IN DIFFERENT FIELDS

ACADEMIA • textbooks and journal articles

BUSINESS • charts (org charts)

COMPUTER GRAPHICS

- technology to create and manipulate visual content

DESIGN • brochures, flyers, posters, websites, and books

EXAMPLES OF GRAPHIC MEDIA

- **POSTER**
- **LOGO**
- **BROCHURE**
- **INFOGRAPHICS**
- **BILLBOARD**
- **MAGAZINES**
- **DIAGRAM**
- **ICONS**
- **MAP**

ELEMENTS OF GRAPHIC DESIGN

- **LINE**
- **COLOR**
- **TEXTURE**
- **SIZE**
- **SPACE**
- **SHAPES**

LINES

- Described as a single point that continues for a distance, or as the connection between two points
- Types of lines are horizontal, vertical, diagonal, curved, thick, thin, dotted dashed, or zigzag

COLOR

- A visual image that uses colors, such as a drawing, illustration, or photograph.
- Color is a powerful tool in graphic design that can convey emotions and messages.

TEXTURE

- The surface quality of an item
- It's how something feels when touched, or looks like it would feel if touched.

SIZE

- The dimensions of an element such as its width, height, or area
- Refers to how little or large an element is in comparison to other things in a design

graphics media

ELEMENTS OF GRAPHIC DESIGN

SPACE

- Refers to the area that a shape or form occupies
- It also refers to the background against which we see the shape or form.
- Space can be defined as positive and negative.
- The **positive space** of a design is the filled space in the design.
- **Negative space** is the background.

SHAPES

- Any 2D area that has an edge
- It's essentially a flat, enclosed form that can be created by lines, curves, or a combination of both

PRINCIPLES OF DESIGN

- **BALANCE**
- **PATTERN**
- **MOVEMENT**
- **EMPHASIS**
- **REPETITION**
- **PROPORTION**
- **RHYTHM**
- **VARIETY**
- **UNITY**
- **CONTRAST**
- **PROXIMITY**
- **HIERARCHY**
- **ALIGNMENT**

BALANCE

- The **distribution of visual weight** within a composition, ensuring that **all elements appear evenly distributed and stable**, preventing any single element from dominating the design
- One should attempt to place visual elements in an aesthetically pleasing arrangement
- Fulfill a purpose or achieve a **particular look and feel**

SYMMETRICAL BALANCE

ASYMMETRICAL BALANCE

RADIAL BALANCE

PATTERN

- Defined as **regular arrangement of repeated same elements** i.e. line, shape, colors over and over again.

MOVEMENT

- Give artists the ability to **lead a viewer's eyes** around an art piece.
- Show **physical action** on a still image

EMPHASIS

- **Humans are naturally distractible**, and our attention can be drawn to many different places at once
- Through emphasis, the eye would know where to land or what information the designer felt was **most important**.

REPETITION

- Pictorial elements are **completely or partially reused**
- Can be the same or similar colors in the design, the use of the same or similar lines, fonts, sizes, and textures, or certain shapes of elements.

PROPORTION

- **Relative size relationship** between different elements within a composition
- How the size of one element compares to another, and how those size relationships contribute to the overall visual harmony of the design
- Creating a pleasing balance by ensuring elements are sized appropriately in relation to each other.

RHYTHM

- A principle that uses repetition to create a sense of movement or pattern.
- Varying the size of objects, shapes, or lines in sequence
- Using a progression of colors from tints to shades

VARIETY

- Disrupt the monotony of a composition
- More engaging to look at
- It keeps the viewer's eye moving around and exploring the different elements

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PRINCIPLES OF DESIGN

UNITY

- How well the elements of a design work together
- Visual elements should have clear relationships with each other in a design.
- Used to create a sense of wholeness in a design by connecting different elements

CONTRAST

- When two or more visual elements in a composition are different
- When elements in a composition differ significantly; the human eye naturally perceives and prioritizes those differences.

PROXIMITY

- The concept of grouping related elements together to create a sense of organization, clarity, and visual connection
- When elements that share a relationship are placed close to each other, the viewer naturally perceives them as a single unit rather than separate pieces of information

HIERARCHY

- The control of visual information in an arrangement or presentation to imply importance.
- Influences the order in which the human eye perceives what it sees.

ALIGNMENT

- The placement of visual elements so they line up in a composition
- We use alignment to organize elements, to group elements, to create balance, to create structure, to create connections between elements

AN ELEMENT OF ART W/ 3 PROPERTIES

HUE OR TINT

- The distinguishing color or then a me of the color, such as red, blue, or yellow

INTENSITY

- The degree of purity, saturation, or strength of a color

VALUE

- The lightness or darkness of a hue

GRAPHICS TECHNIQUE

- Refer to the various methods and tools used in visual design to create, enhance, and manipulate images, layouts, and typography for communication and artistic expression.
- Help designers convey messages effectively and aesthetically across different media

EXAMPLES:

- **TRADITIONAL MARK MAKING**
- **DIGITAL MARK MAKING**
- **DIGITAL FLAT GRAPHICS**
- **3D MODELLING**
- **ANIMATION**

MARK MAKING

- The act of creating a visual mark or trace on a surface using a variety of tools and techniques.
- Fundamental element of art that forms the basis for drawing, painting, sculpture, and many other artistic expressions.

EXAMPLES:

- **STIPPLING**
- **CROSS-HATCHING**
- **SCRIBBLING**

DIGITAL FLAT GRAPHICS

- Refer to the various tools used in visual design to create, enhance, and manipulate images, layouts, and typography for communication and artistic expression.

3D MODELLING

- The process of creating a 3D model of an object or surface using specialized software. It's a computer graphics technique that involves manipulating vertices, edges, and polygons in a simulated 3D space.

ANIMATION

- The process of creating the illusion of movement by rapidly displaying a series of still images, called frames.

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PIXEL

- A **tiny dot or square** that represents a single color in a digital image.
- Pixels are the **building blocks of digital images**, and they combine to form the images we see on screens.

RESOLUTION

- Refers to the sharpness and detail of the image
- It's essentially the number of pixels that make up the image.

RGB COLOR MODEL

- A system that uses red, green, and blue light to create colors for digital displays

CATEGORIES OF DIGITAL GRAPHICS

- **BITMAP**
- **VECTOR**
- **MEDIA**
- **ANIMATED**

BITMAP

- A way of representing an image using a grid of pixels, where each pixel is assigned a specific color.
- Imagine a grid of tiny squares, each square representing a pixel.
- Each pixel has a color value associated with it, and the combination of all these color values creates the image.
- Always orientated horizontally and vertically

MONOCHROME

- Consisting of or using only one color
- Used to describe things that are black and white, but it can also refer to things that are shades of a single color, like different shades of gray or blue.

VECTOR

- Defined by mathematical equations that describe lines, curves, and shapes.
- Equations specify the points, lines, and curves
- This mathematical representation is what allows vector images to be scaled to any size without losing quality.

VECTOR DRAWINGS

- A way to describe any type of vector art creation
- Means creating points, lines, and shapes in vector software using your mouse or a drawing tablet and stylus
- Used to describe the process of rendering an existing piece of vector art within your software program.
- Refers to a digital artist using vector software to create art.

IMAGE FILE FORMATS

JPEG (JOINT PHOTOGRAPHIC EXPERTS GROUP)

- .jpg, .jpeg
- A widely used format for photographs and web images.
- Uses lossy compression, reducing file size while sacrificing some quality.

PNG (PORTABLE NETWORK GRAPHICS)

- .png
- Supports lossless compression and transparency (alpha channel).
- Ideal for graphics, logos, and images that require a transparent background.

GIF (GRAPHICS INTERCHANGE FORMAT)

- .gif
- Supports animation and transparency.
- Limited to 256 colors, making it less suitable for high-quality images.

SVG (SCALABLE VECTOR GRAPHICS)

- .svg
- XML-based format used for web graphics and illustrations.
- Can be scaled without losing quality.

PDF (PORTABLE DOCUMENT FORMAT)

- .pdf
- Supports both vector and raster images.
- Commonly used for sharing and printing documents with embedded graphics.

WEBP (WEB PICTURE FORMAT)

- .webp
- A modern format developed by Google.
- Supports both lossy and lossless compression, with smaller file sizes than JPEG and PNG.

AI (ADOBE ILLUSTRATOR)

- .ai
- Proprietary format used by Adobe Illustrator for vector-based designs.

graphics media

TYPES OF GRAPHICS DESIGN

WEB DESIGN

- Web design is quickly becoming one of the most valuable skills for creative professionals.
- This field of graphic design blends technology and the traditional principles of composition and form to create digital media.

UI AND INTERACTIVE DESIGN

- UI (user interface) and interactive design have become increasingly valuable skills.
- This field focuses on bridging the gap between complex computing and the user experience through visually pleasing forms and layouts.

ADVERTISING AND MARKETING DESIGN

- Focuses on creating visuals that promote brands, products, or services to attract customers.
- Includes advertisements, brochures, social media graphics, banners, and promotional materials.

MOTION GRAPHICS AND ANIMATION

- Involves the creation of moving visuals to convey messages dynamically.
- Includes animated logos, explainer videos, and visual effects (VFX).

PACKAGING DESIGN

- The process of designing product packaging to attract customers and protect the product. Includes box designs, labels, and custom packaging solutions.

GAME DESIGN

- The visual and artistic development of video games, including characters, environments, and UI elements.
- Includes concept art, 3D modeling, and user experience (UX) design.

ILLUSTRATION

- The creation of original artwork for various media, often hand-drawn or digitally painted.
- Used in books, advertisements, merchandise, and entertainment.

PUBLICATION AND TYPOGRAPHIC DESIGN

- Involves designing layouts for print and digital publications, focusing on readability and aesthetics.
- Includes books, magazines, newspapers, and typography-focused designs.

video media

VIDEO

- A video is an excellent tool for delivering multimedia.
- It is a program, movie, or visual media product featuring moving images, with or without audio, that is recorded and saved digitally or videocassette.

MEDIA

- Media is a broad term referring to the means of communication.

VIDEO MEDIA

- Refers to any form of media that delivers content through video format, combining moving images and sound

EVOLUTION OF VIDEO MEDIA

ANALOG VIDEO (1920s - 1940s)

MECHANICAL TELEVISION SYSTEM

- The earliest video systems used mechanical scanning devices, such as the **Nipkow disk**, to transmit moving images.

ELECTRONIC TELEVISION

- The invention of the **cathode-ray tube (CRT)** enabled fully electronic television broadcasts, replacing mechanical systems.

BROADCAST TELEVISION (1940s - 1970s)

- The introduction of **NTSC, PAL, and SECAM** standards allowed television signals to be transmitted over the airwaves, making video accessible to households worldwide.
- Early broadcasts were in black and white

CONSUMER HOME VIDEO (1970s - 1990s)

- The rise of home video began with **VHS and Betamax**, enabling consumers to record and watch content on demand.
- The **LaserDisc (1978)** offered better quality but remained niche. By the 1990s, **DVDs** replaced VHS with digital quality and greater storage capacity.

DIGITAL REVOLUTION (1990s - 2000s)

- The shift from analog to digital transformed video technology.
- The introduction of MiniDV, MPEG compression, and HD broadcasting allowed for higher resolution and easier editing.
- DVDs and later Blu-ray (2006) became the standard for home media, while digital television replaced analog broadcasts.

INTERNET AND STREAMING ERA (2010s - PRESENT)

- The rise of the internet changed video consumption with different platforms
- Streaming replaced physical media
- Smartphones and digital cameras introduced HD, 4K, and even 8K video recording
- Modern codecs like H.265 and AV1 optimize streaming quality

HOW DOES VIDEO WORKS

VIDEO CAPTURE PROCESS

- When light reflected from an object passes through a camera lens, it is converted into an electronic signal by a special sensor called a **Charge-Coupled Device (CCD)**.

ANALOG VS. DIGITAL VIDEO

ANALOG VIDEO

- Resolution measured in horizontal scan lines where each horizontal line represents continuous color and brightness values

DIGITAL VIDEO

- Uses discrete values to represent color and light, similar to an audio signal. Digital video signals consist of RGB (Red, Green, Blue) values for each pixel, making them more precise and versatile.

CONVERSION FROM ANALOG TO DIGITAL

- The process of converting analog video into digital format involves breaking the video frame into data packets.

video media

BROADCAST VIDEO STANDARDS

NATIONAL TELEVISION SYSTEM COMMITTEE (NTSC)

- Standard broadcasting system
- Korea, Japan, most of North America, parts of South America, and the Caribbean Islands
- 1940 by the Federal Communications Commission in the United States
- Nationwide standard for black-and-white TV transmission

PHASE ALTERNATING LINE (PAL)

- The analog video format used in television transmission
- Most of Europe (except France, Bulgaria, and some other countries where SECAM is used), Russia, Yugoslavia, and some Asian, African, and South American countries
- Developed in Germany by Walter Bruch and first introduced in 1967

DIFFERENCE BETWEEN NTSC AND PAL

- **NTSC** - United States
- **PAL** - Europe, Australia, and other parts of Asia

ELECTRICAL DIFFERENCES

- The format difference starts with the electrical power behind each.

RESOLUTION QUALITY

- PAL produces fewer frames per second than NTSC, but it produces more lines of resolution.
- **NTSC** - 525 lines of resolution
- **PAL** - 625 lines of resolution

Système Électronique Pour Couleur Avec Mémoire or Sequential Colour with Memory

- An analog television system that transmits colour information sequentially, using lines of colour information.
- Eliminate the colour artefacts found on systems using the NTSC standard.
- It uses a 625-line vertical, 50-line horizontal display.

ADVANCED TELEVISION SYSTEMS COMMITTEE

- A standard organization that promotes the establishment of technical standards for all aspects of advanced television systems.
- Based in Washington, D.C.

DIGITAL TELEVISION

- An advanced broadcasting technology that has transformed your television viewing experience.
- Offer television with better picture and sound quality
- It also offers multiple programming choices, called **multi-casting**, and interactive capabilities.

ANALOGUE VIDEO

DEFINITION

- Video information stored using television video signals, film videotape, or other non-computer media.
- Each frame is represented by a fluctuating voltage signal known as an analogue wave or composite video.

VIDEO COMPONENTS

- Brightness
- Color
- Synchronization

PROBLEMS

- Color blending issues
- Low clarity
- High generation loss
- Difficult to edit

HOW ANALOGUE VIDEO SYSTEMS PROCESS AND STORE COLOR INFORMATION

- CCD (Charge-Coupled Device) signals are split into three color channels: **red, green, and blue**.
- Video quality depends on how well these signals are **separated** and **transmitted**.
- Best quality: Each color channel has its own **conductor**

video media

HOW ANALOGUE VIDEO SYSTEMS PROCESS AND STORE COLOR INFORMATION

- **S-Video:** Transmits luminance and chrominance through two separate channels.
- **Composite Signals:** All color info is **combined** and sent through a **single cable**. Harder to separate colors and accurately reproduce image quality. (lowest quality)

DIGITAL VIDEO

DEFINITION

- It is the digitization of analogue video signals into numerical format, creating the illusion of full motion by displaying a rapid sequence of changing images on a display device.

CONVERSION PROCESS

- Uses an **ADC (Analogue to Digital Converter)** to convert analogue signals to digital format.
- Uses a **DAC (Digital to Analogue Converter)** to output digital video to analogue equipment.

PLAYBACK AND REQUIREMENTS

- Computer's monitor without special hardware

CONVERSION PROCESS

- Computer with FireWire connection.
- Fast processor, plenty of RAM, and a big hard disk.

DIGITAL VIDEO STANDARDS

ADVANCED TELEVISION SYSTEMS COMMITTEE (ATSC)

- It's the technical standard for Digital TV (DTV) approved by the FCC (Federal Communications Commission).
- Uses MPEG-2 compression to reduce data size.
- Achieves a 50-to-1 reduction by only transmitting parts of the screen that have not changed since the previous frame.

INTEGRATED SERVICES DIGITAL BROADCASTING (ISDB)

- Can be obtained for free at the Japanese organization DIBEG website and at ARIB.
- Standards include ISDB-T (terrestrial), ISDB-S (satellite), and ISDB-C (cable). Uses MPEG-2 or H.264 transport stream structures.
- Supports high-definition television (HDTV) and standard definition television (SDTV).

ENHANCED DEFINITION TELEVISION SYSTEMS (EDTV)

- Modified conventional systems to improve vertical and horizontal resolutions.
- Known as a bridge between standard TV and HDTV by increasing scanning lines from 525 to 1050.

DIGITIZING VIDEO

- Digital video combines graphics and audio features to create dynamic multimedia content.
- The video is simply moving pictures and can be edited easily.
- Digitized video files can be extremely large.
- Digital video is often used to capture content from movies and tv to be used in multimedia.
- A video source (e.g., camera, VCR, or TV) is connected to a video capture card in a computer.
- Analog signals are converted to digital files, including audio.

VIDEO DISPLAYS

INTERLACED SCAN

- The standard for old analog televisions (like CRT TVs)
- It was used to save bandwidth by only showing half of the image at a time (odd lines first, then even lines).

PROGRESSIVE SCAN

- The standard for modern digital displays
- It displays the entire frame at once, making it sharper and better for motion.

Video media

FILE SIZE FORMATS

FRAME RATE

- It refers to the number of individual frames or images displayed per second in a video or animation.
- Animation is an illusion caused by the rapid display of still images.
- 30 fps but 15 fps is acceptable for tv and movies

IMAGE SIZE

- It refers to the dimensions of an image, usually measured in pixels (width x height).
- It determines how much detail the image can show and how large it appears on a screen or when printed.

COLOR DEPTH

- It refers to the number of bits used to represent the color of a single pixel in an image or video.
- It determines how many different colors can be displayed.
- Higher color depth means more vibrant and accurate colors.

SETUP RECORDING VIDEOS

- Fast Processors
- Ample RAM
- Fast and Big Hard Disks
- FireWire or USB
- Adequate Screen Space
- External Speakers

VIDEO EDITING

- The process of manipulating and arranging video clips to create a polished final product.
- It includes cutting, merging, enhancing etc.
- There are two types of video editing: **Linear** and **Non-Linear**.

LINEAR EDITING

- An older method of editing where footage is edited in a sequence from start to finish.
- It's done by copying sections of video from one tape to another
- Commonly used in the days of analog editing.
- It plays end to end in one direction.

NON-LINEAR EDITING

- A modern digital editing process where footage can be accessed and edited in any order
- It refers to the editing of disk-based digital video.
- Editors can work on any part of the video at any time
- Uses advanced editing software
- Allows advanced effects, transitions, and easy revisions.

SPECIAL EDITING

- These are techniques used to enhance or alter video content, creating visuals that are impossible or impractical to capture during filming.
- CGI, green screen compositing, motion graphics, animations, etc.

TRANSITIONS

- Are visual effects used to smoothly move from one video clip to another.
- They enhance storytelling by creating seamless scene changes, indicating shifts in time, mood, or location.
- Such as fading, wiping, splatters, scrolling, stipple and many more

SUPERIMPOSING

- A video editing technique where one image, video, or graphic is layered over another
- Uses of Superimposing: Text & Graphics, Compositing, Visual Effects, Picture-in-Picture (PIP)

VISUAL COMPRESSION

- The process of reducing video file size by eliminating redundant data, making it easier to store, share, and stream.
- There are two types of video compression: **Lossless** and **Lossy**.

LOSSLESS COMPRESSION

- Preserves the exact image throughout the compression and decompression process.

Video media

LOSSY COMPRESSION

- Eliminates some of the data in the image and therefore provides greater compression ratios than lossless compression.

CODEC (LOSSY)

- Eliminates some of the data in the image and therefore provides greater compression ratios than lossless compression.

SPATIAL COMPRESSION

- Compresses video data by reducing redundancy within each individual frame (image data).
- Compression is done by identifying and removing repetitive or similar data within the same frame.

TEMPORAL COMPRESSION

- Compresses video data by analyzing similarities between successive frames over time.
- Stores only the differences between frames instead of the entire frame, reducing file size significantly.

VIDEO PRODUCTION PROCESS

- **PRE-PRODUCTION PROCESS**
- **PRODUCTION PROCESS**
- **POST-PRODUCTION PROCESS**

PRE-PRODUCTION PROCESS

- The creative and logistical process of planning the elements needed to produce a film, TV show, play, or any other creative performance
- Includes:

Video strategy/goals
Budget/scope
Project timeline
Script creation
Talent/characters
Production team/equipment needs
Location scouting

STEPS OF PRE-PRODUCTION

1. Scripting
2. Legal and Budgets
3. Creative Planning
4. Logistics

SCRIPTING

- a crucial part of the multimedia production process

STEPS IN WRITING A SCRIPT

1. Write a video brief first.
2. Turn your message into a story.
3. Speak in a language your audience understands.
4. Keep it short and sweet.
5. Don't just use words.
6. Do several script readings.
7. Tweak, re-tweak, and check against your brief.

PRODUCTION PROCESS

- The actual recording and capturing of the video

ASPECTS OF PRODUCTION

- **the camera**
- **shot types**
- **camera movement**
- **lighting**
- **audio/sound**

THE CAMERA

- Compact Camcorders
- Compact Still Cameras
- Iphone and Iphone
- Action Cameras
- DJI Pocket 2

SHOT TYPES

- Defined by how close the camera is to the subject that's being filmed.

CAMERA MOVEMENT

- Refers to the way a camera shifts to visually narrate and shape a viewer's perspective of a scene.

video media

- 7 basic camera movements:
ZOOM, TILT, PAN, DOLLY, PEDESTAL, TRUCK, RACK FOCUS

LIGHTING

- Fundamental to film because it creates a visual mood, atmosphere, and sense of meaning for the audience

AUDIO / SOUND

- Sound is a really important part of filmmaking. You need to choose the sounds you want and cut out the ones you don't want.
- The best way to do this is to use a separate microphone, not the one in the camera.

POST-PRODUCTION PROCESS

- The third and final stage of video creation
- It's a detailed process that involves many different services, skill sets, and types of professional software, all working together to create a final version of your video or film.

LOGGING THE INTERVIEWS

PRODUCING THE FINAL STORY

MUSIC SELECTION

VIDEO EDITING

REVIEWS AND APPROVALS

FINAL DELIVERY

WHAT'S INVOLVED?

- Editing
- Color Grading & Correction
- Visual Effects & Animation
- Sound Renovation
- Foley Recording
- Script Recording
- Sound Recording
- Sound Mixing & Editing
- Soundtrack & Music

WHO'S INVOLVED?

- Music
- Editors
- Visual Effects Artists
- Animators
- Composers
- Sound Mixers & Editors
- Sound Recordists
- Color Graders
- Foley Artists
- Voiceover Artists
- Actors

5 STAGES OF POST-PRODUCTION

1. Logging Footage
2. Assembling Footage
3. Rough Cut
4. Fine Cut
5. Final Cut

1. LOGGING FOOTAGE

- This is the process of reviewing all the recorded video clips and organizing them
- It helps editors quickly find the necessary footage during editing.

2. ASSEMBLING FOOTAGE

- The selected clips are placed in a rough order or sequence according to the script or storyline
- This is the initial step in structuring the video and ensuring continuity.

3. ROUGH CUT

- A preliminary version of the video is created by trimming unnecessary parts, arranging scenes properly, and aligning audio.
- Focuses on the flow of the story but may not have final effects

4. FINE CUT

- This is a refined version of the rough cut where the pacing, transitions, and overall structure are perfected.
- Minor details like shot timing, sound synchronization, and smoother edits are adjusted.

5. FINAL CUT

- The completed version of the video
- This is the version ready for distribution or presentation.

Video media

PEOPLE INVOLVED IN THE POST-PRODUCTION PROCESS

FOLEY ARTISTS

- Responsible for recreating sounds for film
- Replace or enhance sounds that may not have been recorded properly on set

MUSIC EDITORS

- Responsible for creating the music for a film
- Might use tracks from established recording artists or compose original music

SOUND MIXERS

- They handle sounds from voices, music, and effects such as footsteps.
- They adjust the volume and balance to ensure clarity.
- Their work ensures that important sounds are clearly heard.

VISUAL EFFECTS ARTISTS

- Create the film's computer-generated effects
- Include background alterations and scene enhancements
- Often used in green screen shooting, makeup effects, and explosions

TYPES OF VIDEO PRODUCTION

Educational Videos

Film and TV Production

Television Commercials

Corporate Videos

Product Videos

Customer Testimonial Videos

Marketing Videos

Wedding Videos

ELEMENTS OF AN EFFECTIVE VIDEO

Length

Sound

Script

Lighting

Content

Editing

Format

Captions

LENGTH - the amount of time occupied by something

SCRIPT - the written text of a play, movie, or broadcast

CONTENT - information made available by a website or other electronic medium

FORMAT - the way in which something is arranged or set out

SOUND - narration and any background music are as clean as possible

LIGHTING - the deliberate use of light to achieve practical or aesthetic effects

EDITING - the process of selecting and preparing written, photographic, visual, audible, or cinematic material used by a person or an entity to convey a message or information

CAPTIONS - a movie or television show subtitle

TYPES OF CAMERA SHOTS AND ANGLES

CLOSE-UP SHOT

- When you want to reveal a subject's emotions and reactions
- Fills your frame with a part of your subject
- A shot taken of a person or object at a close range

EXTREME CLOSE-UP

- Fill a frame with your subject
- It often shows eyes, mouths and gun triggers.
- Smaller objects get great detail and are the focal point
- The subject is tightly framed in a relatively large scale

Video media

MEDIUM SHOT

- One of the most common camera shots
- Emphasizes more of your subject while keeping their surroundings visible
- Also called waist shot
- Captured at a medium distance from the subject
- Used for back and forth dialogue

EXTREME LONG SHOT

- Make your subject appear small against their location
- Make your subject feel distant or unfamiliar
- Focus is the surrounding area

LONG SHOT

- Same idea of the long shot, but a bit closer
- Your subject is a person then his or her whole body will be in view — but not filling the shot
- Shows the full length of the subject while also including a large amount of the surrounding area

HIGH-ANGLE SHOT

- The camera points down at your subject
- Creates a feeling of inferiority, or “looking down” on your subject
- Used to make the subject below seem vulnerable

LOW-ANGLE SHOT

- Frames the subject from a low camera height looking up at them
- Emphasize power dynamics between characters
- The camera is positioned low on the vertical axis, and looks up at an object above

OVER-THE-SHOULDER

- Is the perspective of the shot
- Shows your subject from behind the shoulder of another character
- It's common in conversation scenes
- Establish eye line of where each character in the scene is looking
- Most commonly framed through a medium or close-up shot

BIRD'S EYE VIEW SHOT

- This is a shot from way up high.
- It establishes a large expanse of scenery.
- The camera is located up above, capturing the action going below.
- Captured using a drone

DUTCH ANGLE OR TILT

- The camera is slanted to one side.
- More of a stylistic approach to cinematography

TOP VIDEO PLAYERS

- VLC Media Player
- PLEX
- KMPlayer
- PotPlayer
- 5KPlayer
- Kodi
- Media Player Classic Home Cinema
- DIVX
- GOM Player
- FX Player

VIDEO FILE FORMATS

MP4

- MPEG-4 Part 14 is the most common type of MP4.
- MPEG-4 (Moving Pictures Experts Group 4)
- Apple's preferred format
- It uses MPEG-4 encoding algorithm to store video, audio, subtitles, and images.
- It offers better quality than some others.
- Widely used for videos posted on YouTube, Facebook, Twitter, and Instagram.

MOV

- QuickTime Movie (MOV) stores high quality video, audio, and effects but results in large file sizes.
- Developed by Apple
- MOV files use MPEG-4 encoding.
- Preferred by Facebook and YouTube
- It works well for TV viewing.

AVI

- Audio Video Interleave
- Developed by Microsoft
- AVI files offer good quality but large file sizes.
- Supported by YouTube and Facebook
- It works well with nearly every web browser on Windows, Mac, and Linux.

AVCHD

- Advanced Video Coding High Definition
- Developed by Sony and Panasonic
- Designed for professional high-definition video recording
- Uses compression to keep file sizes small without sacrificing quality
- Works well for HD video playback

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MKV

- Matroska Video
- Developed in Russia, Matroska Multimedia Container
- A flexible format that is free and open-source
- Widely supported by many devices but not all web browsers
- Best used on a tv a computer using an open source media player like VLC or Miro

WEBM / HTML 5

- A video format made especially for videos embedded on the internet.
- They are smaller files, so they load quickly and work well on personal browsers.

MPEG - 2

- If you want to burn your video to a DVD, MPEG-2 with an H.264 codec is a way to go.

FLV, F4V, AND SWF

- Flash video formats FLV, F4V, and SWF
- FLV (Flash Video)
- F4V (Flash MP4 Video)
- SWF (Small Web Format or ShockWave Flash)
- Historically common formats for streaming content online
- Flash is not supported by iOS

WMV

- Windows Media Video
- A compression format developed by Microsoft
- It is primarily designed for streaming and storing videos on Windows-based devices.
- Offers high compression with good quality
- Used in storing videos, online video streaming, sharing videos over the Internet, and playing videos on Windows Media Player

ADVANTAGES OF USING VIDEO

Captures interest

Increase retention

Clarifies complex physical actions and relationships

Can incorporate other media

DISADVANTAGES OF USING VIDEO

Expensive to produce

Requires extensive memory and storage

Requires special equipment

Does not effectively illustrate abstract concepts and static